

NICHOLAS HO

Software Engineer | Backend & Full Stack

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SUMMARY

Computer Science student (Binus University Global Class, GPA 3.83) and backend-focused full-stack engineer, shipping production software since 2023. Currently a Backend Software Engineering Intern at SIRCLO. Built enterprise logistics and finance systems in C#.NET at Galva Group, and solo-built payment and marketplace systems in Go and Java with idempotency, double-entry ledgers, and event-driven services, proven by concurrency tests. Published ML researcher; led student teams of up to 6 engineers.

WORK EXPERIENCE

Software Engineering Intern (Backend) | SIRCLO

Aug 2026 - Present

- Top committer on the Project Portfolio Dashboard, an internal app that turns engineering's GitHub Projects boards into progress, risk, velocity, and QA views. Built and deployed it with TypeScript, Next.js, Postgres, Drizzle, Docker Compose, and Caddy.
- Gathered requirements with the VP, Engineering and QA Managers, and PMs; co-authored the technical spec and ERD.
- Built a GitHub GraphQL sync job and an org webhook receiver that logs every status change and its author, plus Google sign-in, per-menu access control, and At Risk email alerts.
- Computed delivery metrics as SQL rollups in Postgres: weighted epic progress, At Risk schedule flags, time-in-status bottlenecks, and QA rejection cycles.

Lead Developer | Nexus Software Agency

Oct 2025 - Present

- Lead development of client landing pages in Next.js and Tailwind CSS, from business requirements to release.

Full Stack Developer (Part-time) | Galva Group

Apr 2024 - Jul 2026

- Built backend logic for an enterprise Warehouse Management System (VB.NET, ASP.NET Web Forms, SQL Server/T-SQL): delivery orders, stock opname, putaway/picking completeness, multi-warehouse hierarchies, and token-authenticated external API integrations.
- Built Kargolo, a logistics admin system for carrier networks and expedition services: React/TypeScript SPA (Redux Toolkit, Zod, MUI) over a .NET 8 Web API using the Repository and Unit of Work patterns with Dapper.
- Built a Project Management System with Next.js, Tailwind CSS, Supabase, and AWS S3.

Full Stack Developer (Freelance) | Galva Group

May 2023 - Aug 2023

- Built the Inventory finance module, which posts balanced double-entry General Ledger journals for multi-currency cash and bank disbursements.
- Designed an API-first C# ASP.NET Web API backend (Repository pattern, Dapper) with TransactionScope cross-database transactions and token auth, and moved the frontend to a Vue.js 3 SPA.

PROJECTS

Agora | C2C Marketplace & Payments Platform (solo build)

Go, Python, Next.js

- Built Go services (PostgreSQL, Redis) for identity, escrow payments, and limited-stock drops: an OAuth 2.0 / OIDC provider written from the RFCs (PKCE, JWKS, TOTP MFA), an idempotent double-entry escrow ledger, and sharded Redis Lua reservations that never oversell. Python services handle risk scoring and AI-assisted listings.
- Published events through a transactional outbox to Redpanda (Kafka API), released the relay as outboxkit, a standalone open-source Go module, and verified it with chaos and k6 load tests: no lost events or double charges.

Switch | Card Payment Authorization Engine (solo build)

Java, Spring Boot

- Built a payment switch with an encrypted card vault, an idempotency layer that runs exactly once under 20 concurrent threads, a rules-based risk engine, and acquirer failover (Resilience4j). Proved it with 104 state-machine tests and 2,000 property-based sequences (jqwik, Testcontainers) that keep the ledger balanced.

Localist | Directory SaaS with Paid Listings (solo build, live)

Laravel, Stripe

- Built and deployed a directory of ~6,100 SEO pages covering 5,800 real businesses, with a Livewire owner portal, Stripe subscriptions (Laravel Cashier), Cloudflare edge caching with targeted purges, and 60 PHPUnit tests.

FAQ Assistant | RAG Document Q&A (Technical Lead, team of 3)

Python, LangChain

- Led Agile sprints and built a local RAG pipeline (LangChain, HuggingFace embeddings, FAISS) that answers questions about uploaded PDFs with exact source excerpts and no external LLM API.

EDUCATION

Computer Science (Global Class) | Binus University | GPA 3.83 / 4.00

2023 - Present

Student Exchange, Computer Science | Sejong University, Seoul | GPA 4.40 / 4.50 (top 2%)

Feb 2026 - Jun 2026

RESEARCH

Machine Learning Algorithms for HIV/AIDS Prediction using Explainable AI

ICORIS 2025, published

- Co-authored; 97% accuracy with Random Forest and XGBoost, with SHAP making predictions interpretable for clinicians.

Hybrid MediaPipe-GRU Architecture for Efficient BISINDO Recognition

Thesis, ongoing

- Lightweight sign-language recognition for budget devices; applied in Signlingo, a gamified web app (Flask, TensorFlow).

SKILLS & CERTIFICATIONS

Programming: Go, Java, Python, TypeScript, JavaScript, C#, SQL, PHP, VB.NET, C

Backend: Spring Boot, ASP.NET Core / .NET 8, Node.js, Express, FastAPI, Flask, Laravel, REST, OAuth 2.0 / OIDC

Data & Messaging: PostgreSQL, SQL Server, MySQL, Redis, Kafka (Redpanda), Supabase, Drizzle, Dapper

DevOps & Testing: Docker, Terraform, GitHub Actions (CI/CD), AWS S3, Cloudflare, OpenTelemetry, JUnit 5, Testcontainers, Playwright, k6

Frontend: React, Next.js, Redux Toolkit, Tailwind CSS, Vue.js, Livewire

AI / ML: Scikit-learn, XGBoost, TensorFlow, LangChain, FAISS, RAG, SHAP, OpenCV, MediaPipe, Pandas

Spoken Languages: English (IELTS Academic 7.5), Indonesian (native)

Certifications: Alibaba Cloud Associate: Cloud Engineer (2025), NVIDIA DLI: Building Conversational AI Applications (2025)